YIFEI SUN

Staff Software Engineer · Ads Platform Infrastructure & Experimentation · 13 Years

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SUMMARY

Staff-level platform engineer with 13 years building high-traffic distributed systems, ads infrastructure, and experimentation platforms in Java. Led cross-team redesign of a high-traffic API serving millions of daily impressions (p99 250ms→50ms, availability 98%→99.9%), validating the migration at 50% production traffic through automated parity testing and controlled rollout. Built an experimentation and evaluation platform with A/B testing, parameter optimization, and automated analysis. Established architecture and observability standards across 10+ teams and 3 departments. Early engineer at Toast through hypergrowth; combines deep hands-on Java engineering with cross-organizational technical leadership.

LEADERSHIP HIGHLIGHTS

- Led cross-team API redesign with controlled A/B rollout at 50% traffic: p99 5× reduction, 99.9% availability
- Drove architecture and observability standards across 10+ teams, 3 departments via RFCs and design docs
- Built experimentation platform with A/B testing, parameter search, and automated evaluation pipelines
- Mentored senior engineers to Principal-level scope; ships 0→1 products fast with measurable impact

Technologies: Java, Spring Boot, GraphQL (Netflix DGS), Go, Python · AWS, Kubernetes, Kafka, Redis, PostgreSQL · Microservices, Event-Driven, CQRS · A/B Testing, Experimentation, Feature Flags · Datadog, Grafana, Distributed Tracing · CI/CD, Docker

INDEPENDENT PROJECT

Experimentation & Evaluation Platform

- Built an end-to-end experimentation system for decision evaluation: A/B simulation, offline backtesting, parameter search (grid & Bayesian optimization), and automated analysis with configurable metrics and thresholds.
- Designed a configurable policy framework with DB-backed configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions — enabling safe, data-driven iteration.
- Architected a hybrid rules + multi-path inference engine over heterogeneous data sources with full observability, telemetry collection, and auditability across the entire experiment lifecycle.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead Oct 2019 – Present

High-Traffic API Modernization — Experiment-Driven Platform Migration

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind the most-visited page (~8,000 rpm, millions of daily impressions); validated the new implementation at 50% production traffic via controlled A/B experiment.
- Introduced a new orchestration service and GraphQL boundary, consolidating 12 downstream integrations into 5 well-defined services, reducing fan-out and simplifying failure modes.
- Built a parity-testing framework that automatically compared new vs. legacy responses across full payload and data-integrity dimensions, enabling safe, data-driven rollout decisions.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)
- Defined architectural standards and observability requirements for new microservices; instrumented services with structured logging, distributed tracing, and SLO-based latency budgets per downstream dependency.

GraphQL Federation — Delivery Lookup DGS Subgraph

- Designed and integrated a Netflix DGS subgraph into a federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies to avoid JIT compilation latency spikes on cold starts.

MCP Server — Agent Orchestration & Tool Calling (0→1)

- Sole backend engineer designing and implementing an MCP server for agent orchestration, exposing platform capabilities as tools callable by AI agents (ChatGPT via OpenAI Apps SDK).

- Delivered a performant implementation iterating on ambiguous requirements; designed for safe, incremental production rollout alongside existing APIs.

Digital Shopping & Commerce Experimentation

- Led backend design for a new digital shopping product launched during the pandemic, using microservice and event-driven architecture for inventory, merchandising, and lead flows.
- Built infrastructure for rapid product experimentation — enabling the company to quickly launch, measure, and safely retire product bets without destabilizing core systems.
- Earlier at CarGurus, built the initial Facebook Ads integration pipeline (MediaMath for budget allocation, impression tracking, and click-through rate calculation).

Cross-Org Platform Leadership & Observability Standards

- Established patterns for API design, observability, caching, and distributed orchestration adopted across multiple teams; authored design docs, RFCs, and decision records driving consensus.
- Drove adoption of structured logging, distributed tracing, and metrics-driven development practices. Mentored senior engineers on system design and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Modernization

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Introduced social login and mobile login; migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Initiative

- Drove early design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth (first customer to hundreds in a few years); designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the basis for future growth.
- Built initial AWS deployment pipelines before a dedicated DevOps team existed; created the first automated testing framework before dedicated QA, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010